



Novel nano material

Researchers have synthesized a nano material with electrical and magnetic properties for use in Quantum Electronics. <https://digit.in/oct18-73-1>



Smallest medical robot

120nm Nano drug delivery robots for Cancer now hold the Guinness World record for smallest medical robot. <https://digit.in/oct18-73-2>

an artificial synapse, that allows for precise control of the electricity flow, similar to the way biological synapses in the human brain regulate the flow of ions. Traditional neuromorphic chips used an amorphous substance as a switching medium between two conductive substances to simulate the synapses. On application of a voltage, the electricity would flow in random and imprecise ways through this switching medium. This meant that an individual synapse would never perform uniformly. The team from MIT used single crystalline silicone, which is a defect free material, where all the individual atoms are neatly ordered. The engineers created a single line defect, a single dislocation, over which the ions could flow predictably. The development was a key step towards realising artificial neural networks. The researchers fabricated a chip with each synapse measuring 25 nanometers across. The new system was put to the test by being trained to recognise handwriting. The hardware was capable of accurately recognising handwriting for 95 percent of the time. Software algorithms that do the same thing have demonstrated an accuracy of 97 percent.

BODY ON A CHIP

Researchers at Harvard Seas and MIT have created various human organs on chips, that can be used for a number of tests. The lung on a chip for example, consists of a tiny groove with vacuum pipes on either side. Within the groove is a membrane that separates the groove into two channels. The top side of the membrane is deposited with lung cells, while the bottom side is deposited with capillary cells. The top channel simulates the air flow, while the bottom channel simulates the blood flow. The groove can expand and contract into the vacuum pipes on either side, simulating the breathing of the human body. Introducing bacteria into the air channel, allows researchers to study lung infections. Similarly, living cells deposited on a chip can simulate the

intestine, the gut, the kidney, the liver, the pancreas, skeletal tissue, the brain and the heart. Connecting up multiple such tiny organs on a chip, allows researchers to create the entire human body on a chip. The chips can be used to develop new drugs and novel treatments. Researchers can study how the drug is distributed through

diovascular system of the human body for example, the heart acts as a “pump” to make the blood flow to all the parts of the body. Engineers from MIT have designed a “tree on a chip” that mimics the passive circulation system of plants. The passive hydraulic system takes advantage of the surface tension of water to transport the fluids.



Credit: Felice Frankel

A body on a chip system that simulates 10 organs

the body, and how it gets metabolised. The effect on other tissues after treating a particular part of the body can also be evaluated. For example, a new drug can be introduced into the guts portion of the chip, mimicking the “swallowing” of a pill. Then the drug can be observed as it travels to the different parts of the body and study the effects. Such chips could also be used to reduce the dependence of pharmaceutical and cosmetics companies on animal testing.

TREE ON A CHIP

Trees and plants have tissues called xylem and phloem. The xylem transports water and other nutrients from the roots to all the parts of plants, while the phloem transports sucrose and other products of photosynthesis, to wherever they are needed within the plant. The system is passive, and has no moving parts. Within the car-

One of the major breakthroughs over previous attempts was the use of sugar, which increases the osmotic pressure of the water. The chip is very simple, it is just two slides of plastic with drilled through channels to represent the xylem and phloem. The xylem channel is filled with water, and the phloem channel is filled with water and sugar. The two slides are separated by a semipermeable membrane. A similar membrane is also

found within plants. A sugar cube was placed on the phloem channel to diffuse the sugar into the system. A pipe connected to a tank was attached to the chip. This simple system managed to maintain a steady water flow for several days. This was a significant improvement over previous attempts that managed to keep the water flowing only for a few minutes. The tiny chip can be used as a miniaturised source of hydraulic power for robots. It is difficult to manufacture the small moving parts for nanoscale robots, and adding components can be difficult and expensive. The tree on a chip design allows small scale hydraulic robots to be built easily, and cheaply. The small scale robots can mimic the actions of their larger cousins, such as the hydraulic robots created at Boston Dynamics. The innovation could allow tiny robots to be powered by “fuel cells” made up of sugar cubes.